All right, good afternoon, recap of last time. So we were talking about the situation where we are sampling data from a normal population, IID random variables X1 through XN from a normal distribution, each with the same mean mu, the same variance in the squared, and the case where those are both unknown, both the parameters of that normal distribution are unknown, which is usually what would happen in practice. Usually we wouldn't have information, we wouldn't have knowledge of one of those parameters. want to estimate them both, or at least we'd have to consider the fact that they're both not no. So in that situation, this is our MLE for the mean, the Mughat point the MLE for the mean, and it's just the sample mean, X bar, and our MLE for the variance in this situation is 1 over n times the sum of the squared deviations around the sample mean.

Deviation meaning xI minus mu hat or x bar.

Square deviations, sum them up, divide by n. That's the MLE for the And these are both functions of the data X, both functions of random data, so they are themselves random variables. Mu hat is a random variable, sigma squared hat is a random variable. So they have some distribution, and in particular, because there's two of them, both involving X, they have a joint distribution, some bi-variant distribution. And so yesterday's, or last Tuesday's lecture, was centered on this situation and specifically the sampling distribution of those two quantities, the sample mean and the and the leaf of the variance. And recall the lecture was centered around this one theorem with three parts, right? So under this setting, the first part says that the sample mean, X bar, or MLE of mu, follows a normal distribution. That normal distribution has mean mu, so the true mean, X bar is centered around the true mean, and the variance of sigma squared over ed. So this tells us the sampling distribution of X bar. For sigma squared hat, the MLE of the variance, we had that this scaled version, So scaling it by n over sigma squared, this scaled version follows a chi-square distribution, n-1 degrees of freedom. So the n-minus one comes from the fact that we're plugging in x-bar here. We're plugging in our estimate of mu, right? We saw in a previous lecture, if we knew mu, if we had the true value from mu, and calculated the squared deviations around that, then this scaled variance estimate would be kai-squared with n degrees of freedom. But we lose one degree of freedom by estimating that mean parameter or plugging in X bar in these square deviations. So that's the second property. And then the third was that these two quantities are independent. So our sample mean X bar and our MLE for the variance, sigma squared hat, are independent of each other. And that was really the point of the lecture last time, right? Just trying to reinforce these facts about the sampling distribution for the MLE of the mean variance under this IID normal sampling. situation. Yeah. So if both the variants have different distributions why is it bivariate? So a bivariate distribution just means a distribution of two things. So together these make up a vector of length two, right? X bar and sigma squared hat are a vector of length two. So any distribution has to vary in two dimensions. It's bivariate. So that's all it's saying. If you're just looking at the distribution of x bar, that's a univariate distribution. If you're just looking at the distribution of sigma squared hat, that's universe distribution, but together they have a joint distribution, although this particular joint distribution, they're independent. So we know we get all the properties of independence. We could write the joint PDF as a product of the individual PDFs because of independent.

So that's an important property here. Okay, so that was our lecture last time, all about the sampling distribution from a normal population, random sample from a normal population. Today we're going to take another detour and talk about or introduce a new distribution. This time the T distribution.

So again, I don't teach the T distribution when I teach stat 131. If you did cover it, then I guess this will be review, but I cover it here because I don't teach that as part of my 131 cloud. But this T distribution are often called the students' T distribution comes up in a lot of places. It's going to be related to our sampling distribution that we talked about last time. It's going to involve the kai square distribution that we've been building up, so we've been

working up to this. But it comes up in different places. We'll use it again next week when we talk about interval estimation. But today we're just going to kind of build the tools around this distribution, working with the T distribution, understanding the T distribution, how we can get random variables that follow this distribution. So I'm going to start with this generative definition. So kind of like the kai square distribution, we had a way to generate from it, right? We could take a bunch of independent, standard normal random variables, square them, add them together. it gave us a way to generate draws from that distribution. Well, this definition is a generative definition. It gives us a way to generate or draw samples from this distribution based on some more primitive random variables. So here, the first primitive random variable is going to be Z, a standard normal random variable, and the second is going to be Y, which will follow a kai square distribution with m degrees of freedom. So we have these two random variables, Z and Y, and they're independent. This is important. Z has to be independent of Y for this construction to work. Then if we construct this new random variable that I'll call T, and we construct it by taking the random variable Z and dividing by the square root of the random variable Y scaled by one over its degrees of freedom of M. It involves the random variable Z and involves the random variable Y. If we create this new random variable T when these conditions are met, this random variable T will follow what's called a T distribution, and this T distribution has one parameter, M the degrees of freedom parameter. So it just inherits that M from whatever kai square was used to make it. But notice we're scaling the kai squared random variable by 1 over m in the construction here. And this is the notation we'll use to denote a random variable that has this T distribution. So the random variable capital T has a T distribution denoted little t sub m or t with m degrees of freedom. And we have this constructive definition now. So we know if we have a way to generate a standard normal random variable, and we have a way to generate a kai square random variable, then we have a way to generate T distributed random variable. So we could do that through simulation if we wanted. Important properties of this distribution, it's a symmetric distribution. So just like the normal distribution, symmetric, this one's symmetric. And it's symmetric about zero. So if it's symmetric about zero, we know that 50% of its probability, mass falls to the right of 0, 50% of its probability mass falls to the left of 0. Again, just like the standard normal distribution.

Moments, so if we call moments from Stat 131, right, this is the definition of a moment. The expectation of the absolute value of the random variable X to the keth power is called the kath moment of the random variable X. We use moments to evaluate things like the mean of the variance. But in this case, we have to be careful. So, And we have to look at our degrees of freedom parameter. And so if we're considering moment K, as long as our degrees of freedom parameter M is greater than whatever moment we're thinking about, then this expectation is finite, that moment exists. But if our degrees of freedom is less than or equal to K, then that expectation is infinite. That moment does not exist. So the moments aren't always defined for the T distribution, like they are for, say, the normal distribution. And so in particular, what are the, implications of this well think about if m is equal to one and we want the first moment one is less than equal to one so that first moment is infinite there's no mean if the t distribution has one degree of freedom there's no mean the mean does not exist yeah question k is the the kth moment so if k is one it's the first moment if k is two it's the second moment it's just an index for whatever moment of interest we might it's just this expectation of the absolute value of the random variable to the kth power right but we use use it to calculate things like the mean and the variance. So the first moment is the mean. We can, you know, a centered version of the second moment is the variance and so on. There's skewness is related to the third moment. Cortosis is related to the fourth. So it's just these expectations of these absolute value of x to the kth power. Because if you take the definition of this expectation, so you integrate this function, absolute value of x to the kth power times the probability density function, that integral is not finite. So by definition, that expectation doesn't exist, or it's infinite. If the expectation was zero, say the first moment, then the mean would just be zero. It would exist. But if that expectation is infinite,

right? It's not finite. We don't have a first moment. So that's the condition is if it's a finite value, if this expectation is some finite value, less than infinity, the moment does exist. But if it's infinite, then it does not exist. So we have a normal random variable. and we're dividing by this square root of y over m, which is going to be positive. So we're dividing by some positive value. But this y over m, it could be less than one or it could be greater than one. So the actual effect is that it's going to inflate the tails relative to the normal distribution. So extreme values are going to be more likely. Values close to zero are going to be less likely. So that's what's going to happen when we scale by that. So be careful if m is one, if we're talking about a, T distribution with one degree of freedom. The mean does not exist. The distribution is still symmetric about zero. It always is, but the mean does not exist. There is no mean, no finite mean.

If m is greater than one, so we have more than one degree of freedom, then the mean does exist and it's zero. Centered around zero in either case, right? In either case, 50% of the probability mass falls to the left, 50% to the right. But the mean, the expectation of the random variable does not exist in the case when m is equal to one or one degree of freedom.

And then again, if m is one or two, the variance does not exist. For any m greater than two, the variance does exist. And it's equal to m over m minus two. The degrees of freedom over the degrees of freedom minus two. So that's the consequence of not always having these finite moments, right? The mean, the variance don't always exist. If we have enough degrees of freedom, they will. Right. But if that degrees of freedom is small one, and we don't have the first moment or the second moment, so on, yeah. And then do you say variance of X by X you mean a T random variable, or a random variable that falls? A random variable that falls a T distribution, yeah. Yeah, if the degrees of freedom is one, the first and second moment don't exist, so there's no variance. And if the degrees of freedom is two, the variance still doesn't exist. But if it's three, then the second moment does exist and the first moment exists. And if it's 3, the variance is 3 over 3 minus 2. Question? Sorry, the random variable X here is the unstandardized G? It's just any random variable that follows a T distribution with M degrees of freedom. And again, I mentioned this, but this T distribution, the shape of it is it's going to have heavier tails than the normal distribution. So if you plotted the probability density function, right, extreme values, values away from zero are going to have higher probability density. values close to zero are going to have lower probability density relative to the normal distribution. But this effect of heavier tails kind of vanishes as the degrees of freedom increases, as M gets bigger. So as M gets bigger, these heavier tails shrink closer and closer to the normal distribution. And in particular, as M goes to infinity, this T distribution actually converges to the standard normal distribution. So the standard normal distribution is a limiting case of this T distribution as the degrees of freedom goes to infinity. And as the degrees of freedom goes to infinity, all the moments exist, just like the standard normal distribution. I'm going to give you the probability density function here, just for completeness, but it's really not that helpful. So this is what it looks like. We have this normalizing constant involving the parameter M, and then we have this term that describes the shape of it. So if you were interested in probabilities involving this T distribution, right, you would integrate this probability density function. That's kind of tedious to do. Again, nowadays, people would use software to do it, or in our case, we're going to use a table to do this. And so if you go to the end of the lecture notes, we have this T table, just like we had for the kai squared. So this is something you'll get on your exam, you'll have access to this T table, you'll have access to the kai squared table, and so you can use it. to find probability is related to the T distribution. Let's talk about how to use this real quick. So it says if X, the random variable X, has a T distribution with M degrees of freedom, that's our parameter. And just like the kai square table we looked at a little bit ago, this table gives the value of little x such that the probability that the random variable big x is less than or equal to little x is equal probability random variable big x less than equal to little x written in green here is equal to p written in blue so if we were to pick one of these rows for example so i'm

going to pick m equals 10 so if we look at this first column we have m we have different values for the degrees of freedom we could pick one of these i'm going to pick 10 this here and let's say i look at this value that i've circled in green we see that value in that row is associated with this probability p of 0. This is saying for a random variable T, I'll say X, for a random variable X, follow the T distribution, 10 degrees of freedom, probability that that random variable X is less than or equal to the green value. We've circled here, 0.879, that probability is what, 0.8, the blue value. So that's how we can read this table. What about the column? Point 5. We don't have a column for 0.5. Why not? 25 is 0. Yeah, it's symmetric about 0. So regardless of the degrees of freedom, whatever it is, we know 50% of the probability masses to the right, 50% to the left. And it is symmetric, which is why we also don't have anything less than 0.5. We could use symmetry to get those values. Say, T distribution looks something like this, pretend that's symmetric, zero in the middle. So this gives us probabilities to the left of some value. So if we had some value X here, this table gives us the probability to the left of x, but what if instead we wanted the probability of the right of x?

Well, that's not what the table gives us, but we can use symmetry.

If we reflect x across zero, we have some value negative x over here, and this table does give us the probability to the left of values, to the left of negative x.

Specifically, we can say probability is less than or equal to negative x, to 1 minus probability that the random variable x is less than equal to positive little x.

One, one minus the area to the left of little x is the green area on the right, and that's equal to the red area on the left. So we don't need a column for 0.5. We don't need a column for anything less than 0.5. We can use symmetry to get there. That's the idea. One other thing I'll point out is if you look at the very last row here, we have degrees of freedom is infinity, right?

This is the limiting case or the standard normal distribution. I don't give you a separate standard normal probability table. You can use this row of the T table for any probabilities involving the standard normal distribute. I would say that, but it's approximates a normal distribution approximates that T distribution pretty well when you start to get, you know, large degrees of freedom, 30, 40, so on. Even 29, 28, it approximated it pretty well. But yeah, that's the idea. As you increase that degrees of freedom, it gets closer and closer to a normal distribution. Or another way of saying that is that a normal distribution is a better and better approximation for that T distribution. If you look at these probabilities here and in 30, there's a little bit of difference, but they're pretty similar, right? Those distributions are pretty similar. Question back here. So you could use software or you could just try to interpolate between the two values. Yeah, so on an exam, if you only have this, you don't have access to software. If it's right between interpolate, if it's pretty close to one value, just use that value. And really there's, you know, infinite numbers of columns that could exist. And I can't give you an infinite table, so I have to give you some finite set of columns. Again, you'll have access to this table on your exam. Yeah, a question. Two up there, but don't we exclude those two? No, the distribution still exist, and you can still calculate probabilities. They just don't have any finite moments. So I can't give you a mean or variance, but I can give you the probability that the random variable falls in some rate. and still do that.

If you're interested at all, M equals 1 is a special case called the Koshi distribution, C-A-U-C-H-Y, special distribution.

Really interesting properties. It comes up in a lot of special cases. If you take the sample average, the variance of that sample average does not decrease as you take more and more samples. It's a very counterintuitive, not what we would expect. So very interesting distribution. But yeah, that's a special case

of the T distribution when m is equal to one.

Yeah. So in this case, there's nothing to estimate, regardless of M. Like, if you know m is 10, there's still nothing to estimate.

There's no other parameters. M is the only parameter here. We're not going to be using this to estimate parameters of the T distribution, but it's going to come up when we talk about interval estimation and stuff. And so the idea is you can have shifted and scaled versions of this.

So take a T distribution that's not mean zero. Yeah, not centered at center it somewhere else and scale it out right it's a sort of a generalized t but just like with the normal distribution we can take a z score take something that has mean mu and variance sigma squared and transform those random variables to something that's standard normal you can sort of do the same thing here right and so that'll come up next week when we talk about interval estimation but yeah we're not necessarily trying to estimate anything here yeah we're going to make probability statements about random variables that you know some transformation of them or function of them have a T distribution. So that was the probability density function. Here I'm just plotting this probability density function for some different values of the degree of freedom. So in red is the normal distribution. That's this one that has the highest peak, right? And it has the smallest tails down here. This green one has the lowest peak, has the highest tails. That's two degrees of freedom. And then in the middle are blue with five degrees of freedom, purple with 30 degrees of freedom. And again, you can see purple, 30 degrees of freedom, and the normal distribution are very, very similar. There's not much practical difference to using one versus the other. But if you know it is T with 30 degrees of freedom, you know, it's still a little better to use that distribution directly rather than a normal. But, yeah, hopefully you can see visually here, this T distribution, particularly for small degrees of freedom like two, has much heavier tails. So the probability of being to the right of, say, three is, it's not something here. huge value, but it's much bigger for this T distribution than tiny little probability for the normal distribution. Heavier tails, so extreme values, values away, far away from zero, are more likely for the T distribution. But these heavier tails kind of vanish as that degrees of freedom increases, and it gets closer and closer to the normal distribution. So how does this relate to what we've been talking about? Well, we've been talking about normal sampling. So this is where it comes up.

So remember we're talking about a random sample, X1, through XN, IID from a normal distribution that has mean, mu, and variance sigma squared. Often we'll calculate the sample mean, call that x bar, and then a little bit different from what we've been talking about. We've been talking about the N-O-E for the variance, right? So we call it sigma-squared hat, and we were dividing by N. This is another estimator for the variance. It's often just called the sample variance, or S-squared. It's the same thing. We're just dividing by n minus 1 here, but it's still the sum of squared deviations around. that sample mean x bar.

So if we're talking about sample mean x bar, and this s squared, our sample variance, and we construct this quantity here.

So it involves our sample size, square root of n, and then we have x bar minus mu, and then we divide by the sample variance or the sample standard deviation, square root of s squared.

This quantity is going to follow a T distribution or have a T distribution, and its degrees of freedom will be n minus one. This is how it relates to normal sampling.

Yeah. So why do we divide by n minus 1 here to get this sample variance? Why do we do it? In this specific case, it's what gives us a T random variable, right? Because the degrees of freedom is n minus 1, so by dividing by n minus 1 here, that gives us a T random variable. In other cases, the MLE we saw is biased.

This we'll talk about next week is actually an unbiased estimate for the variance. That's another reason you might do that. Either one is an estimator for the variance, though. just kind of your own preferences on which one you might choose. And in fact, as N gets very large, they're more or less the same. That difference becomes smaller and smaller. But yeah, here it's because it gives us a T. U is just a random variable constructed with this formula. I'm just giving it to you here. I'm just telling you that if you constructed a random variable this way, that random variable would follow a T distribution, and that T distribution would have n minus 1 degrees of freedom. What we'll do with that is gonna come next week, but for now, just recognize that this thing will have a T distribution. So if you divided like the numerator of you by like sigma squared hat instead of S squared, you wouldn't get like a T distribution with n degrees of freedom. Yeah, because even in sigma squared hat, remember we're losing a degree of freedom by estimating X bar. We're going to get n degrees of freedom. We're not going to get a kai squared. Remember it's a it's constructed using kai square. We're not going to have a kai squared with n degrees of freedom. We lose one. And I think this will help to just show the proof for this. understand where this is coming from.

Okay, so what do we know? We know currently that this random variable that I'm going to call z, but if you took square root of n, x bar, this meal over the square root, sigma squared.

So the true variance, instead of the sample variance, what distribution does this have?

Standard normal? Just follow the standard normal distribution. I think this is the first question of your homework. We've also done it somewhere else in the notes if you were observant. Okay, so this is one thing we know. This z constructed using the true variance. follows a standard normal distribution. We also know from our last lecture, this other random variable I'm going to call y, sum bar squared deviations around the sample mean x bar over the variance sigma squared. The same thing, n minus 1, s squared, or sigma squared. If you take s squared and you multiply by n minus 1, you just get the sum of the squared deviations. N minus 1 s squared over sigma squared follows what distribution. Does anybody remember? kai squared with how many degrees of freedom, n minus 1. So we're starting to get the ingredients we need for a T random variable. We have n, we have kai squared. We also show that these two things are independent last time. That was our third property. And so now we can start to construct a T random variable based on our generative definition that we gave it early. We're going to have a standard normal in the numerator. We're going to have a kye squared divided by its degrees of freedom, a square root, in the denominator. Those are the ingredients. Instruct a T random variable. Now, new random variable T is Z for the square root of Y, divided by its n plus one degrees of freedom. We know that this will have a T distribution, and it will have n minus one degrees of freedom. Yep. We'll still do problem solving, but we're going to do so relying on certain properties that need to be shown to do them. So we need to be comfortable with these properties before we can use them to solve problem. If I ask you something about the sampling distribution in a normal population, you need to know what that sampling distribution is first, which is why we had the last lecture, right? And that relied on the kai squared distribution, which is why we had that lecture. And next week we're going to talk about interval estimation, which is going to use the T distribution. And so the proofs are there to sort of convince you that these properties are true. Then you can use those properties to solve problems when they arise. Many different distributions come up in confidence intervals. T distribution comes up a lot when you talk about confidence intervals under normal sampling, which is one of the main cases of interest, and so that's why we're spending a lot of time on it. But yeah, there's other distributions, and we'll look at other examples too, but there's other distributions that will come up in confidence intervals. As we've mentioned, the normal distribution is by far the most widely used distribution statistics. It comes up in many, many different examples. That is due to the central limit theorem, which basically says, you know, if you're adding up or averaging a bunch of independent things, right, that converges to a normal

distribution, while a lot of things in life are, you know, the sum of many individual decisions, like you take a test, you answer many, many questions, maybe they're independent, right? And so your average, your test score might be normally distributed, or something like your height might be influenced by many, many different genetic factors. And so that's, you know, That's why we see the normal distribution come up so much. But definitely not the only distribution. There's other distributions that are useful as well. So here we have constructed this Z, constructed this kai squared random variable Y with n minus 1 degrees of freedom. And based on our definition of the T distribution, we know that this thing will follow a T distribution with n minus 1 degrees of freedom. And we can rearrange this a little bit now. So we can plug in our expression for z into the numerator. So the numerator is square root of n x bar minus mu, square root of sigma squared. The denominator, s squared over sigma squared. It will be divided by n minus 1, canceled out. And then we see that the square root of sigma squareds cancel out. So this reduces to square root of n, x bar minus mu, over the square root of s squared. But it is standard normal random variable divided by square root of a kai square random variable over its degrees of freedom, both independent. And so it must have a T distribution. This will follow a T distribution, N minus 1 degrees of freedom. So that's why we used the sample variance rather than the MLE of the variance. By dividing by n minus 1 here, we got something that follows a T distribution. Say this is using the definition of the T distribution. Yeah, so we construct a variance estimate using these sum of squared deviations. right?

If they're around the true mean, right, if we used mu instead of x bar, then the scaled version of this would have a kai square distribution with n degrees of freedom. But if we don't know mu and we instead plug in x bar, lose a degree of freedom.

So that's the difference. If we plug in the true value mu, when we calculate this, if we had access to that, then it would be end. Yeah, so we showed it in the last lecture, but intuitively the mean, the sample mean x bar, is the sum of all your x is divided by n. So if I give you the first n minus one values of x and I give you the sample mean, there's only one value that the last x could be.

It has no ability to vary. There's no degree of freedom for that last observation.

So by estimating x bar, you're reducing the degrees of free.

It's the intuitive explanation. We showed it mathematically. Yeah, we showed it for zes, specifically. That was the first theorem that we proved in the last book. X bar is the MLE for me. So if we didn't know Mew, we would plug in our estimate for it, which is X bar. This is an unbiased estimate for sigma squared, if not the MLE for sigma squared. But the degree of freedom that we're losing is not from this N minus 1, right? Because when we talk about the kai square distribution, it's unbiased, but that's an expectation. On average, over many, many, many samples, they'll average to the true value. But still, if I give you X bar and I give you the first N minus one observations, there's only one value that that last observation can take. It's only the one that makes them sum to N times X bar. It's the only value it can take. It can't be anything else. So this relates to our normal sampling. So now if we have our normal sampling setup and we ever see this, you can use the theorem and you know now it follows a T distribution with N minus one degrees of freedom. And it will come up next week when we talk about intervals. Let's just do a couple examples. with this T distribution. So we're going to assume we have a random variable T that follows a T distribution with 5 degrees of freedom, m equal to 5, and we're just asked to find this probability. The probability is random variables between negative 2 and 2. We know we can use our table to do this, but let's just get it into a form where we can look up a value in our table. We know our table is probabilities to the left of a value and that value has to be between 0.5, or sorry, above the mean, right, so positive, or above the center, zero, so positive. So we want to get this expression into something that we can look up, or maybe multiple things that we can look up in our T table. We have the

probability, T is between negative 2, positive 2. We know we can split this into two probabilities. Probability, T is less than or equal to 2, which we could look up, minus the probability T is less than equal to negative 2, if we can't look up directly. Everybody comfortable with that. But we know that the probability that t is less than or equal to negative two is one minus the probability that t is less than equal to two. Probability two minus one probability t is less than or equal to two. That was our symmetry result that we showed with our total. We can do that. So we have two of these probability t less than equal to two terms, and we have a negative one. So this is just two, probability, T2 minus 1, there's only one value we need to look up in our table here. The probability of the random variable T is less than equal to 2, and we're talking about a T distribution with 5 degrees of freedom, so we know which row of our table we need to look at. We go to our table, looking at this row, where m is equal to 5, and we're looking for some value here, equal to 2 or close to 2, and we see this one here is 2.015, close enough, and that probability associated with it is 0.95. So going back to our problem now, this is roughly 2.95 minus 1 or 0.9. There's about a 90% probability with a T random variable with 5 degrees of freedom falls between negative 2 and 2. Next example, now we have a random sample, x1 through x10, from a normal distribution with mean 50 and variance 4, and we're asked to find the distribution of this expression, square root of 10 times the sample mean x bar, minus 50 over the sample standard deviation or the square root of the sample variance. You know from our theorem here that this quantity u follows a t distribution and use that that u which is the square root of n or square of 10 x bar minus mu is 50 in this case over square root of s squared this is just square root of n times x bar minus mu the square of s squared right same thing from our theorem this is specifically for n equals 10 a sample size of 10.

So directly by that theorem, by theorem I think, yep, theorem 2.1, this random variable U will follow a T distribution, n minus 1 degrees of freedom, or specifically for this case, T distribution with 9 degrees of freedom.

So that's just using our theorem directly. Now we have a random sample of size 26 of these rainfall measurements, X1 through XN, taken from a normal distribution, unknown mean mu, unknown variance sigma squared. We're interested in x bar and specifically how far it'll be from the true mean. We'd like it to be close to the true mean, but it's random, so it could be more or less far away from the true mean. And we're going to use this quantity that we know follows a T distribution with n minus 1 degrees of freedom or theorem, and we're asked to find the 90th percentile of x bar terms of mu and s squared. So let's start with what we know. We know this random variable U is going to follow a T distribution, and we're asked to find percentiles, and we have a T table. So let's start there. Probability, U is less than equal C. If we set that to 0.9, then C is the 90th percentile of the random variable U. What we're asked for, but it's a starting point. Okay, so let's find C. Let's find this 90th percentile of U. And we know the sample size, N, is 26, so how many degrees of freedom for the T distribution that U follows? 25 and minus 1. U follows a T distribution 25 degrees of freedom. And so we're looking for the 90th percentile of a T distribution, 25 degrees of freedom. Highlight this column because we're looking for the 90th percentile, and we said 25 degrees of freedom. I'll highlight that row, and at the intersection we have 1.316. We found that C, the 90th percentile of this T distribution is 1.316. We're actually asked about probability or a percentile involving x bar, not u, but use a function of x bar, so that can help us. We know probability of the random variable u is less than equal to 1.316, that probability is 0.9. And now let's just plug in our expression for you. Plug it into this probability statement. This is the same as saying probability square root of n, square root of 26 in our example, x bar minus mu, over the square root of s squared. That's just u, probability that's less than equal to 1.316. That probability is 0.9. And now we're just going to start rearranging this inequality to isolate x bar. So this is the same as probability x bar minus mu is less than or equal to 1.316. Square root of s squared over the square root 26. That probability is equal to 0.9. More step, mu over to the other side. We have the probability x bar less than equal to mu 1.316 to the square root of s squared with a square root of 26. We know this

probability is 0.9. We knew U was a random variable with a T distribution. So we were able to find its 90th percentile C, 1.316, and we just took this inequality and rearranged it to get x bar on the left hand side. And now we see this is the 90th percentile of x bar. The probability that the random variable x bar is equal to this value is 0.9. That's the definition of the 90th percentile. And so this is the 90th percentile of x bar in terms of mu and s squared. This idea of taking a probability statement and rearranging the inequality to isolate what we're looking for. We did this when we did our kai squared lecture. We did it again here. It's going to come up again next week when we talk about interoperative. estimation.

Get very familiar with this idea of taking a probability statement and rearranging the inequality inside.

This says where. Any questions here? C stands for? Yeah, it's just a constant. We didn't know the value, right? It's the 90th percentile. We didn't know it initially, but we were able to 316.

look it up in our table and find that it's this constant 1. Okay. Last example here. Suppose we observe sample size n equals 20 cheese chunks with lactic acid concentrations these random variables x1 x-n and these come from a normal distribution again with unknown mean mu and variance single squared find the value c should actually say that this probability is equal to 0.95 so go ahead and add that the problem's complete we want to find the value c that this probability statement is pretty similar to the last example and do something pretty similar start with a probability state for a random variable that we do know and rearrange it into one that involves x bar here. What random variable do we know probabilities for? We have a probability statement involving x bar, but we don't know the distribution of x bar. So what is a distribution that we do know that we could look up probabilities for? What's a random variable that does have a known distribution for us? T distribution, specifically this random variable that we can call you, you can call it T if you want. Call it whatever, but it's calculated in this way. square root of n times x bar minus mu over the square root of s squared. We know that that thing follows a t distribution with n minus one degrees of freedom. So we're going to start there. We know this random variable u, square of n, square root of 20 in this case, x bar this mu, the square root of s squared. We know that this will follow a t distribution with how many degrees of freedom. 15, n minus one. So that gives us a starting point. Talking about probability or a percentile, a 95th percent, So let's go ahead and find the 95th percentile of U because we can find that. In other words, the probability that you is less than equal to some constant. I'm going to call it C0. Call it whatever you want though, C0. The probability that the random variable is less than equal to that constant, 0.95. That makes that constant the 95th percentile. So we're going to go ahead and use our T-table to find what that constant is. Find C-0. Yeah. Yeah, we're just going to find this percentile first and we'll use it later when we go to that expression. So this C is not necessarily our C0. That's why I'm calling it C0 and not. It's just some constant that we're going to find right now. You can. I'm going to show how we get there. Yeah. So we have a T distribution with 19 degrees of freedom, and we're looking for the 95th percentile. This column, this row gives us this value 1.729. That's our C0. That's that constant that we're looking for. C0 v.1.729. And now let's do what we did last time. We have a probability statement involving U. Plug in our expression for U, square root of 20, X bar, this U, mu, over the square of S squared. This is U. This is how we construct U, and we know the probability that this random variable less than or equal to C0 is 0.95, or rearranging probability x bar is less than equal to mu, c0, squared of s squared over the square of 20. that is 0.95. So we're almost there, right? We have mu plus c0 times squared of s squared over square of 20. On our right hand side, we just want mu plus some constant c times square of s squared, right? So we see that this c0 over the square root of 20 is our constant. Probability x bar less than or equal to mu plus that constant c0, 1.7 to 9, square of 20, square root of s squared, which is the same as x bar less than equal to mu 0.3866 squared of s squared. This probability is 0.95. So our constant C is just this value here. 3866.

0. Questions here. Sorry, which stuff? Okay, let's put away electronics. Take out a piece of paper. You're going to do an extra credit activity. Same deal as always. You can work together. You can use any paper copies of notes. You can use any physical version of the textbook. So you'll have 20 minutes starting now. And then when you're done, just write your name. And then your answer, just the letter is what I need. Circle it and put it in here. 29. I do not have. I guess they're unknown, right? Doesn't really matter. Let's see, what do we learn today? Okay, so first of all, we need to make it, we need to standardize it, because our X's are not standardized. We want it to be like Z, you know what I mean? Well, I mean, I'm assuming that the extra credit is about this, right? Yeah, but if you go ahead to like you, right? Like, you don't have to, like, end-notes that view is sort of n, so that's all the T-distribution, you know, that, but I think it's zero to for that. Oh, I see, yeah, yeah, yeah, you're right, okay. So it's N minus one degree to three-down and it's five. It's going to be a two-distribution and four degrees of freedom. But, um, sorry, what? I mean, you're going to follow it in T distribution? Oh, sure. Okay, yeah, sure. That's, that's our, that's our, that's my, But we should just write this down, right? So we should just write this down. I should write this down. We said N is five, right? We don't know what the bar is. We don't know what new is. We don't know what new is. It's just a normal distribution. It's a, it's a key distribution, right? It's a normal. Wouldn't he be, wouldn't UBT? Because this is like the open-sere? It's like you're not necessarily that is what it's past. Look, they are, he's asking that. You think this is like value? Yeah, yeah. But we have a really, it is. Oh, I see. Yeah. But I'm thinking if we square view, right? Is that what kind of, that, I think it's a lot. I think it's a standard normal. I think it's very good. I feel like to because the perspective because it would be a standard normal because it would be a standard normal because it would be well finding how I think, you know, yeah. It's like a distribution, that it's always constant. I think, so, you know, if you squared, you get something very similar to normal, but it's times n, because you get x bar minus mu over the variance times n, right? And remember, I think you were saying, like, if you have c times a normal random variable, then, like, won't the mean be, like, the mean will be, like, n times new, and then the variance would be like one over n squared. Sorry, can I see my binder? Yeah. Because you're making you think. It's great. It doesn't is. That's a joke. Oh, let's see, um. But normal of a number of zero. Just by the gardener. I think it's five. Or no, it's four. Right. No one is five. It's U is the T. Sorry, just the C is a four, whatever. The distribution of the year is T and four. The variance is n plus square. Yes, because that's the most. Variants plus is 10. Yeah, because the variance is normally C squared square. So where we're in the form of the variance? Are you? I don't know. Uh, I don't know. There's probably a bit. But like the form of the area of the type of. And then the same place. Right here. So we indicated Oh, the whole thing. Yeah. Oh, yeah. Okay, I was talking about X-Ware. Okay. I don't think I'm going to be in the word. Yeah, X2. I was talking about that. Oh, I see. Okay. Well. We know the bear. We know the bear. It's one. The var is in any word. anywhere? What's like the variance you can multiply it like a normal variable by C? No, I didn't have like 1 over c squared? The variance times? You can just like take the... I have to use for... Yeah, yeah, it's time 1 over 2, right? Or the time? Let me look. One over c squares from the variance, which is, let's just assume it's one. It's going to be 1 over 25 times the variance 1 over 25, and then... That's the variance. And then expected value of you is, 10 10 minutes. 10 minutes. 10 minutes. I think you're going to be high square. Wait, wait, wait, but this is very, we're looking for the expectation, right? So we're looking for the expectation, right? Exactly. Which is not working, right? We're trying to find an expectation. The expectation of square is open, we're at it, it won't get, like a second, is squared. Oh no, you're so right, yeah, yeah. Oh, cool. Okay. What's the expected value of high squared? For real. No joke. You're right. Right. Here. Remember this? Oh. Okay. Okay, so much. Wait, so, So, um, oh, because A is 2, and then M is 5. So when K is 2, it's equal to the variance, of you, you know what I mean? Wait, so m is 5, m is 4, and then it's greater than 2, then it's less than infinity. Yes. Okay, oh, and then now it's going to go ahead. Is it m over it? For m greater than 2, the variance, the variance of u. Oh, it's two. Oh, I see.

Oh, yeah, yeah. Okay, I think, I mean, if you got it in. So you mentioned that we're looking for the next of view square, right? So anyway, we're looking for a second point. And equal, line of one. And we're looking for the very moment, which is the variance of view. The variant is the second moment. We're looking for a second moment, which is the variant of you. But I got it. Oh, I tell you're saying, I was going to say, I was going to split, and I was like, what are you talking about, girl? Well, in this case, it's the second one. Yeah, that's the first one we'll do. I wonder if there's any other. Okay, but given the committee, if they expect what I is, you're right? It's, you could be a question of all on the solution. That's what I'm going to be ready. I was thinking. I was keeping up. I was pretty. That's right. Yeah. Oh. And there is. And there is. I don't know. So they're looking for the second square. And we're going to have one. They don't have much. They don't have much. It's a lot. there's a very in the form of room. I know. I do which is the same the, And you can't. And you get out. There's questions like to see in. I know. Okay, two minutes. Okay, two minutes. Okay, two minutes. Has everyone turned theirs in? The last call. Okay, so suppose we have a random sample, size 5, from a normal distribution with mean, mu, invariance, and squared, looking at this quantity, U here, we're asked what's the expectation of U squared. We know the distribution of U from our lecture today, from our theorem, U, that was a T distribution with n minus 1 degrees of freedom.

That was given straight out of the lecture, that was our theorem, or in this specific case, U, T, 4 degrees of freedom. Hopefully we remember from Stat 131 that the second moment here is related to the variance, specifically the definition of the variance of a random variable U is the expectation of U minus the square, the expectation of U. But we know the T distribution, at least the T4 distribution, has expectation zero. So this second term is zero, the mean zero distribution. So the variance of U is just the expectation U squared of U is zero. And again, we know from lecture the variance of a T distributed random variable, when the variance exists, is M over M minus 2, 4 over 2, 2 in our case. And that is the expectation of U squared, or the answer is B. I'll stick around for a bit if there's questions. Otherwise, have a nice weekend. I will see you next week to talk about interval estimation.